Mahiyan Rahman

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EDUCATION

University of Texas at Dallas — **Expected August 2027**

B.S. in Computer Engineering — Richardson, Texas

- Relevant Coursework: Computer Architecture, Embedded Systems, Electronic Circuits, Digital Circuits, Electrical Network Analysis, Signals & Systems, Electronic Devices, Advanced Engineering Math, VLSI, Operating Systems

TECHNICAL SKILLS

Languages: SystemVerilog/Verilog, C/C++, Rust, Python, MATLAB, RISC-V Assembly, TCL
Hardware Design: Vivado, Cadence (Virtuoso, Xcelium, Genus, Jasper, Innovus), OpenLane2, UVM, STA
Computer Architecture: Tomasulo's Algorithm, Out-of-Order Execution, Superscalar Dispatch, Branch Prediction, CUDA
Protocols & Interfaces: AXI4, UART, SPI, I2C, GPIO, ROS2, microROS
Lab & Tools: Ubuntu, Git, Oscilloscope, Multimeter, Logic Analyzer, Analog Discovery 2, LTSpice, Altium, KiCAD

EXPERIENCE

Independent Study Researcher - Parallel Processor Architecture — **August 2026 - Present**

UTD - Advised by Prof. Swartz — Richardson, TX

- Investigating GPU and parallel processor microarchitecture through a year-long independent study in SystemVerilog RTL design under faculty supervision

Embedded Systems Member — **August 2025 - Present**

UTD Comet Robotics - Solis Rover Project — Richardson, TX

- Deployed microROS on an RP2350 Pico 2W over a ROS2 bridge to build a sensor evaluation pipeline for the club's rover
- Diagnosed MOSFET driver faults across a 9-channel payload board and designed a sequencing FSM for bring-up verification
- Developed C++ firmware for Arduino & RP2350 targets with I2C drivers & sensor abstraction layers for modular integration

Roblox Studio Tutor — **January 2024 - December 2024**

Freelance — Remote

- Designed personalized curricula for 15+ students, adapting to individual skill levels to accelerate project completion
- Taught OOP, multithreading, client-server architecture, state replication, and vector math within a live scripting environment
- Guided 5+ students through system design and debugging to ship functional games, emphasizing modular decomposition

PROJECTS

MNIST FPGA Inference Accelerator | SystemVerilog, PyTorch, OpenCV, Python, Vivado, Linux — **August 2026 - Present**

- Training and INT8 quantizing a CNN in PyTorch for MNIST digit classification with weights exported for FPGA development
- Implementing a fixed-point inference engine in SystemVerilog with pipelined convolution, pooling, and fully connected layers using INT8 MAC units and layer-wise requantization

UART SoC Peripheral | Verilog, Cadence (Virtuoso, Xcelium, Genus, Jasper, Innovus), UVM, Linux — **August 2026 - Present**

- Implemented a 16550-compatible UART transceiver in Verilog with UVM functional verification in Xcelium and Jasper
- Designing a custom CMOS standard cell library in Virtuoso with transistor-level schematic, SPICE simulation, and DRC/LVS verified physical layout
- Synthesizing through Genus and Innovus with custom Liberty-characterized cells

Scalar2 | SystemVerilog, Vivado, RTL Design, FPGA, Linux, TCL — **March 2026 - August 2026**

- Designed a 2-wide superscalar OOO processor with Tomasulo scheduling, ROB, RAT, and PRF for dynamic register renaming and precise 2-wide in-order commit on a custom 32-bit ISA
- Designed a Load-Store Queue (LSQ) with conservative store-to-load forwarding for precise memory dependency resolution
- Authored a self-checking testbench against a golden reference model to validate hazard resolution, branch misprediction recovery, and memory ordering correctness
- Implemented a 2-bit saturating counter branch predictor to reduce control hazard penalties and improve IPC
- Achieved 1.25 average IPC (of 2.0 max) via 6-port CDB and 80-85% branch prediction accuracy, verified in simulation

CERTIFICATES & AWARDS

- Altium Designer Essentials: Mastering the Fundamentals
- Dean's List Spring 2024: West Texas A&M University